He-Chen Weng

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Research Interests

Scattering Amplitudes, Perturbative Bootstrap, Cluster Algebras, Non-perturbative Bootstrap

Education

- PhD in Physics, *Brown University* — Sept 2023 - Present
  Advisor: Professor Marcus Spradlin
- BS in Physics, *National Taiwan University* — Sept 2017 - Jun 2022
- BS in Electrical Engineering, *National Taiwan University* — Sept 2017 - Jun 2022

Publications

1. L.-Y. Chiang, Y.-t. Huang and **H.-C. Weng**, *Cornering the Heterotic String*, arXiv:2610.12318 [hep-th]
2. L. Ferro, T. Łukowski, S. Paranjape, L. Ren, M. Spradlin, A. Volovich, **H.-C. Weng** and Y.-Q. Zhang, *de Sitter Wavefunction from Quadrangular Polylogarithms: Polygon Graphs*, arXiv:2610.06772 [hep-th]
3. L. Ferro, T. Łukowski, L. Ren, M. Spradlin, A. Volovich, **H.-C. Weng** and Y.-Q. Zhang, *de Sitter Wavefunction from Quadrangular Polylogarithms: Chain Graphs*, arXiv:2605.06542 [hep-th]
4. S. Paranjape, M. Skowronek, M. Spradlin, A. Volovich and **H.-C. Weng**, *Landau Analysis of One-Cycle Negative Geometries*, arXiv:2604.22683 [hep-th]
5. S. Paranjape, M. Skowronek, M. Spradlin, A. Volovich and **H.-C. Weng**, *Cluster Bootstrap for Cosmological Correlators*, JHEP **09**, 121 (2026), doi:10.1007/JHEP09(2026)121, arXiv:2603.08670 [hep-th]
6. S. Paranjape, M. Spradlin, A. Volovich and **H.-C. Weng**, *Split-Helicity Tree Amplitudes and Flag Cluster Algebras*, JHEP **05**, 130 (2026), doi:10.1007/JHEP05(2026)130, arXiv:2512.09704 [hep-th]
7. A. Pokraka, M. Spradlin, A. Volovich and **H.-C. Weng**, *Symbol Alphabets in QCD and Flag Cluster Algebras*, JHEP **03**, 120 (2026), doi:10.1007/JHEP03(2026)120, arXiv:2506.11895 [hep-th]
8. R. Bhardwaj, M. Spradlin, A. Volovich and **H.-C. Weng**, *Unitarity of bespoke amplitudes*, Phys. Rev. D **110**, no.10, 106016 (2024), doi:10.1103/PhysRevD.110.106016, arXiv:2406.04410 [hep-th]
9. L.-Y. Chiang, Y.-t. Huang and **H.-C. Weng**, *Bootstrapping string theory EFT*, JHEP **05**, 289 (2024), doi:10.1007/JHEP05(2024)289, arXiv:2310.10710 [hep-th]
10. L.-Y. Chiang, T.-C. Huang, Y.-t. Huang, W. Li, L. Rodina and **H.-C. Weng**, *The geometry of the modular bootstrap*, JHEP **02**, 209 (2024), doi:10.1007/JHEP02(2024)209, arXiv:2308.11692 [hep-th]
11. L.-Y. Chiang, Y.-t. Huang, L. Rodina and **H.-C. Weng**, *De-projecting the EFThedron*, JHEP **05**, 102 (2024), doi:10.1007/JHEP05(2024)102, arXiv:2204.07140 [hep-th]
12. L.-Y. Chiang, Y.-t. Huang, W. Li, L. Rodina and **H.-C. Weng**, *(Non)-projective bounds on gravitational EFT*, arXiv:2201.07177 [hep-th]
13. L.-Y. Chiang, Y.-t. Huang, W. Li, L. Rodina and **H.-C. Weng**, *Into the EFThedron and UV constraints from IR consistency*, JHEP **03**, 063 (2022), doi:10.1007/JHEP03(2022)063, arXiv:2105.02862 [hep-th]

Academic Honors

1. Physics Dissertation Fellowship — 2026 - 2027
2. KITP Graduate Fellowship — Winter/Spring 2027
   Mentor: Nathaniel Craig

Seminar Talks

1. QFT Group Seminar, Max Planck Institute for Physics — Dec 2025
   Seminar talk: *Split-Helicity Tree Amplitudes and Flag Cluster Algebras*

Short Talks and Posters

1. Prospects in Theoretical Physics 2026 Summer School — Jul 2026
   Short talk: *De Sitter Wavefunction from Quadrangular Polylogarithms: Chains*
2. Amplitudes 2026 Conference — Jun 2026
   Short talk: *De Sitter Wavefunction from Quadrangular Polylogarithms: Chains*
3. Symbology@15 Conference — Dec 2025
   Short talk: *Split-Helicity Tree Amplitudes and Flag Cluster Algebras*
4. Michigan Interdisciplinary Meeting on Amplitudes: Bridges between Physics & Mathematics — Oct 2025
   Short talk: *The cluster adjacency of split-helicity Yang-Mills*
5. Amplitudes 2025 Conference — Jun 2025
   Poster: *Symbol Alphabets in QCD and Flag Cluster Algebras*
6. Amplitudes 2024 Summer School — Jun 2024
   Poster: *Unitarity of bespoke amplitudes*
7. Amplitudes 2024 Conference — Jun 2024
   Short talk: *Unitarity of bespoke amplitudes*

Selected Conferences and Workshops Participation

1. Amplitudes 2026 School — Jul 2026
2. Lancefest Workshop — Jun 2026
3. Max Planck-IAS-NTU Center Kick-off Conference — Mar 2026
4. From Analyticity to Phenomenology Workshop — Jan 2026
5. Symbology@15 Conference — Dec 2025
6. Amplitudes Afterparty in Taiwan — Jun 2025
7. Amplitudes 2025 Summer School — Jun 2025
8. Summer School in Total Positivity and Quantum Field Theory — Jun 2025
9. QCD Meets Gravity 2024 — Dec 2024
10. Amplituhedra, Cluster Algebras, and Positive Geometry Conference — May 2024
11. Mathematical Aspects of Scattering Amplitudes Program — Apr - May 2024
12. Kickoff Workshop for Simons Collaboration in Celestial Holography — Oct 2023

Academic Appointments

1. Research Assistant, PIs: Prof. Spradlin & Prof. Volovich — Summer 2023; Summer 2025 - Present
2. Teaching Assistant, PHYS2060 - Quantum Mechanics — Spring 2025
3. Teaching Assistant, PHYS0040 - Basic Physics — Fall 2024
4. Teaching Assistant, PHYS2060 - Quantum Mechanics — Spring 2024
5. Teaching Assistant, PHYS1530 - Thermodynamics and Statistical Mechanics — Fall 2023